



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
REVIEW TEST
14-03-2021

JEE(Main + Advanced) : NURTURE COURSE (PHASE : TEPs)

PAPER-1

Time : 3 Hours

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks : 18)

- This section contains **SIX** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. The moment of inertia of a hollow cylinder of mass M and inner radius R_1 and outer radius R_2 about its central axis is :

(A) $\frac{1}{2}M(R_2^2 - R_1^2)$ (B) $\frac{1}{2}M(R_1^2 + R_2^2)$ (C) $M(R_1^2 - R_2^2)$ (D) $\frac{1}{2}M(R_2 - R_1)^2$

2. By varying the voltage applied to the kettle, you can change power consumption P . Depending on the P of kettle, water can be heated to different maximum temperatures. This dependence is shown in table.

Table

Power P (in Watt)	0	100	200	300
Temperature T (in $^{\circ}\text{C}$)	20	40	60	80

If the power consumption is 400W

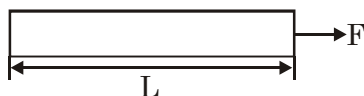
- (A) The water will come to the boiling point and start boiling.
- (B) The water will reach boiling point but will not boil.
- (C) The water will not reach boiling point but start boiling.
- (D) The water will neither reach boiling point nor start boiling.

Space for Rough Work

3. Substance breaks down under a stress of 10^5 Pa. If the density of the wire is 2×10^3 kg/m³, find the minimum length of the wire which will break under its own weight ($g = 10$ m/s²)
- (A) 10 m (B) 2.5 m (C) 4 m (D) 5 m
4. A bullet is fired on a fixed target. It penetrates inside the target through distance $d = 4$ cm and then stops, mass of bullet is $m = 1$ kg and of target is $M = 4$ kg. Now identical bullet moving with same velocity is fired on identical target which is placed at rest on frictionless horizontal surface. Then find the distance (in mm) to which the bullet will penetrate inside the target ?
- (A) 36 mm (B) 42 mm (C) 32 mm (D) 24 mm
-

Space for Rough Work

5. A rod of length L is pulled with a force F as shown on a smooth horizontal surface. If A is the area of cross-section and Y the Young's modulus of the material of the rod, the elastic potential energy stored in the rod is :-



- (A) $\frac{F^2 L}{3YA}$ (B) $\frac{F^2 L}{2YA}$ (C) $\frac{F^2 L}{6YA}$ (D) $\frac{F^2 L}{YA}$
6. ABC is a triangular framework of three uniform rods each of mass m and length 2ℓ . It is free to rotate in its own plane about a smooth horizontal axis through A which is perpendicular to ABC. If it is released from rest when AB is horizontal and C is above AB, find the maximum velocity of C in the subsequent motion.

- (A) $\sqrt{6g\ell}$ (B) $\sqrt{4\sqrt{3}g\ell}$ (C) $\sqrt{2\sqrt{3}g\ell}$ (D) none of these

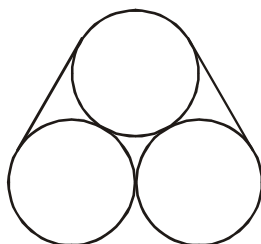
Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 24)

- This section contains **SIX** questions.
 - Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
 - For each question, choose the correct option(s) to answer the question.
 - Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
 - **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.
-
7. A small sphere of radius 25 mm with a surface temperature of 327°C is located at the geometric centre of a large sphere of inside radius 125 mm with an inner surface temperature of 27°C . Both the bodies behave like black bodies. Choose the correct statement(s)
(A) The net rate of heat exchange between the two spheres is 40 W approx.
(B) The wavelength corresponding to the maximum emissive power for the smaller sphere is almost 20 times that of the larger sphere.
(C) Almost 4% of radiation emitted by the larger sphere falls on the smaller sphere
(D) The rate of energy radiated by the inner sphere is 55 W approx.
8. In a long pipe filled with water, we made a partition using a cork 1 cm thick. The partition divides the pipe into two parts. If the water temperatures in the parts of the pipe differ by 1 degree, the flow heat rate through the partition is 2 J / s. Now we add one more partition to the right using cork 2 cm thick. In between the partitions water is filled up. To the left of this region we will maintain the water temperature $+50^{\circ}\text{C}$, to the right - the temperature $+20^{\circ}\text{C}$. Determine the steady state temperature T of the water in the gap. Determine the heat flux Q through each partition. The thermal conductivity of the pipe walls is negligible small.
(A) $Q_{\text{left}} = 10 \text{ W}$ (B) $Q_{\text{right}} = 20 \text{ W}$
(C) $T = 40^{\circ}\text{C}$ (D) $T = 25^{\circ}\text{C}$

Space for Rough Work

9. Three smooth cylinders of radius a are tied together by a circular ring shaped rubber band whose radius in unstretched state is a . The Young's modulus of rubber is Y and the cross-section area is A . $\sqrt{A} \ll a$.

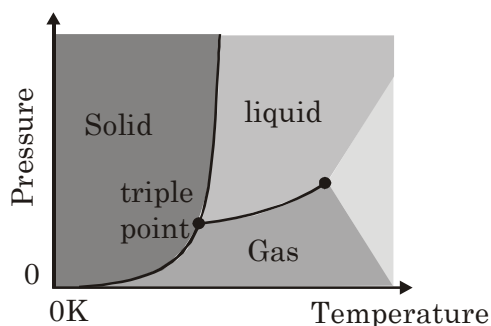


Select the correct statement(s) :

- (A) The tension in the rubber band is $\frac{2YA}{\pi}$
- (B) The normal force between any two cylinders is $\frac{3YA}{\pi}$
- (C) The force exerted by rubber band on any cylinder is $\frac{3\sqrt{3}YA}{\pi}$
- (D) The normal force between any two cylinders is $\frac{2YA}{\pi}$

Space for Rough Work

10. Phase diagram for CO_2 is shown here. The triple point is at 5 atm and -55°C

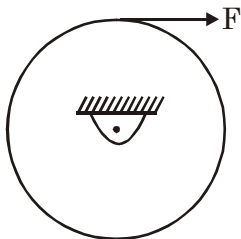


Which of the statement(s) can be definitely concluded from the above phase diagram?

- (A) The melting point of CO_2 increases with increase in pressure.
- (B) The boiling point of CO_2 increases with increase in pressure.
- (C) If we have a sample of CO_2 at atmospheric pressure, it can be in liquid state.
- (D) Suppose that Dry ice (solid CO_2) is existing at room temperature. It can be melted by decreasing pressure on it.

Space for Rough Work

11. A disc of mass M and radius R is hinged on a horizontal smooth surface at its centre. When a force F is applied on the disc tangentially, angular acceleration of the disc is found to be α . About what another point can the disc be hinged so that the angular acceleration of the disc remains same. Also find the initial reaction on the hinge in both the cases (F remains same in magnitude and direction).



- (A) At a distance of $\frac{R}{2}$ from the centre on the other side of the hinge.
- (B) At a distance of $\frac{R}{3}$ from the centre on the other side of the hinge.
- (C) In the first case, reaction by the hinge is F in magnitude.
- (D) In the second case, reaction by the hinge is zero.
12. A smooth sphere A of mass m collides elastically with an equal sphere B at rest. The velocity of A before collision is 8 m/s in a direction making 60° with the line of centres at the time of impact.
- (A) The sphere A comes to rest after collision.
- (B) The sphere B will move with a speed of 8 m/s after collision.
- (C) The directions of motion A and B after collision are at right angles.
- (D) The speed of B after collision is 4 m/s .

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

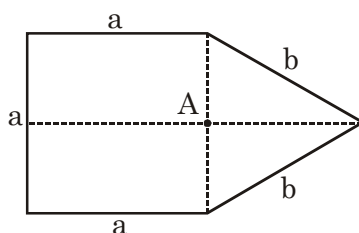
+					-				
●	●	○	○	○	●	●	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	●
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	●	⑤	⑤	⑤	●	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- There are two concentric spherical shells of radii 10 cm and 20 cm. Both of them radiate like black bodies. When the space between them is evacuated and the temperatures are 300 K and 301 K, their rate of heat exchange is W. Now the space is between them is filled with a material of thermal conductivity K such the rate of heat exchange is still W. What is the value of K (in W/mK). (Take : $\sigma = \frac{17}{3} \times 10^{-8} \text{ W/m}^2\text{K}^4$)
- An automobile has a mass of 1500 kg and its aluminium brakes have a total mass of 6 kg. Assume that all the mechanical energy dissipates in form of heat in Aluminium brakes and no energy is lost to the surroundings. If the automobile is travelling at 25 m/s and is stopped. The Aluminium brakes get heated up. How many maximum numbers of times can this automobile be stopped this way before the brakes start melting? Assume that the Latent heat capacity of Al is $4 \times 10^5 \text{ J/Kg}$ and its melting point is 600°C above the temperature at present. The specific heat capacity of Al is $900 \text{ J/kg}^\circ\text{C}$.

Space for Rough Work

3. A cart with a mass of 2000 g has four wheels, each of 6 cm radius and 150 g mass. Calculate the linear acceleration (in m/s^2) of the cart when a force of 0.6 N is exerted upon it. Assume that the wheels are like discs and are rolling purely.
4. Ram acquired a very high-quality thermos (a vessel that eliminates the heat exchange of contents with the environment) with a capacity of 55 cc, the heat capacity of the walls of which is 42 J / K. Initially the temperature of the walls of the empty thermos is 20°C (same as that of the room). Ram successively pours 1 g into a thermos water at a temperature of 1°C , then 2 g of water at 2°C , then 3 g of water at 3°C ... and so on until the filling of the thermos. What will be the steady temperature (in $^\circ\text{C}$) of the contents of the thermos? Water is assumed to have a constant density of 1 gm/cc. ($S_w = 4200 \text{ J/kg K}$)
5. 36 kcal are spent in heating a copper block which weighs 1 kg and is at 0°C initially. What is the final volume of the block (in cc)? $\gamma_{\text{copper}} = 5 \times 10^{-5}/^\circ\text{C}$, ρ_{copper} at $0^\circ\text{C} = 9 \text{ gm/cc}$, specific heat of copper = 0.09 cal/gm $^\circ\text{C}$.
6. The mass center of the homogeneous plate shown in the figure is in point A. What is the ratio b/a ?

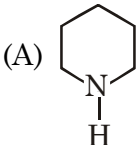
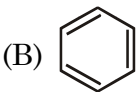
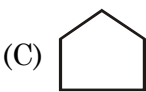
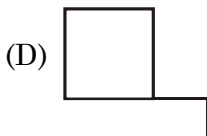


Space for Rough Work

PART-2 : CHEMISTRY

SECTION-I(i) : (Maximum Marks : 18)

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- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. If IUPAC name of an element is "Unununium" then **CORRECT** statement regarding element is :
 (A) It is inner transition element
 (B) It belongs to 8th period in periodic table
 (C) It is transition element
 (D) It is non-transition element
2. In which of the following set of elements first element having higher electronegativity than second element on Pauling scale?
 (A) Ga, In (B) C, N (C) Tl, In (D) Rb, Sr
3. Which of the following compound will be most basic in aqueous phase?
 (A) $\ddot{\text{N}}\text{H}_3$ (B) $\text{CH}_3-\ddot{\text{N}}\text{H}_2$ (C) $\text{CH}_3-\ddot{\text{N}}\text{H}-\text{CH}_3$ (D) $\text{CH}_3-\ddot{\text{N}}(\text{CH}_3)_2$
4. Compound which is heterocyclic -
 (A)  (B)  (C)  (D) 
5. At 298 K, bond dissociation enthalpies of C-C and C=C are 330 and 540 KJ/mol, respectively. If the enthalpy of polymerisation per mole of polyethene $[(\text{C}_2\text{H}_4)_n(\text{g})]$ from ethene gas at 298 K is - 6MJ, the value of 'n' is :-
 (A) 5 (B) 50 (C) 500 (D) 200
6. An equimolar mixture of $\text{Na}_2\text{C}_2\text{O}_4$ and $\text{KHC}_2\text{O}_4 \cdot 3\text{H}_2\text{C}_2\text{O}_4$ required V_1 litre 0.1M KMnO_4 in acidic medium for complete oxidation. The same amount of mixture required V_2 litre of 0.2M NaOH for complete neutralization. What is the ratio of $V_1 : V_2$
 (A) 4 : 7 (B) 10 : 7 (C) 8 : 7 (D) 2 : 7

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 24)

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7. Liquids 'A' and 'B' form an ideal solution. The vapour pressure of solution at 313K is

$$P [\text{in cm Hg}] = 60 + 40 x$$

where 'x' is the mole-fraction of liquid 'A' in the liquid solution at equilibrium. If 'y' represents the mole-fraction of 'A' in the vapours in equilibrium with the liquid solution at 313K, the **CORRECT** information(s) related with the solution is/are

(A) $P_A^0 = 100 \text{ cm Hg}$

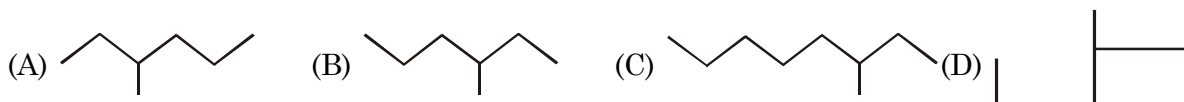
(B) $P_B^0 = 100 \text{ cm Hg}$

(C) If $x = 0.5$, then $y = 0.625$

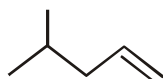
(D) If $y = 0.5$, then $P = 75 \text{ cm Hg}$

Space for Rough Work

8. The reaction : $A(s) \rightleftharpoons 2B(g)$ is in equilibrium at 4 atm and 27°C . If the volume of system is increased at constant temperature, then at new equilibrium, relative to initial equilibrium :-
- (A) Moles of A(s) decreases (B) Moles of B(g) increases
(C) Molar concentration of A(s) decreases (D) Molar concentration of B(g) decreases
9. Which of the following electronic configuration belongs to f-block element?
- (A) $[\text{Xe}] 4f^2 6s^2$ (B) $[\text{Xe}] 5d^1 6s^2$
(C) $[\text{Rn}] 6d^2 7s^2$ (D) $[\text{Rn}] 6d^1 7s^2$
10. Select **CORRECT** statement(s) for overlapping $2p_x + 2p_x(\text{I})$, $2p_y + 2p_y(\text{II})$ (INA = internuclear axis)
- (A) If INA is x, then strength of I > II (B) If INA is y, then strength of I < II
(C) If INA is z, then strength of I = II (D) If INA is x, then strength of I < II
11. Which of the following compounds has same number of carbon in parent chain ?



12. Which of the following IUPAC name is/are incorrect?



- (A) 4-Methylpent-1-ene (B) 2-Methylpent-1-ene
(C) 2-Methylpentane (D) 4-Methylpent-1-yne

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

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For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

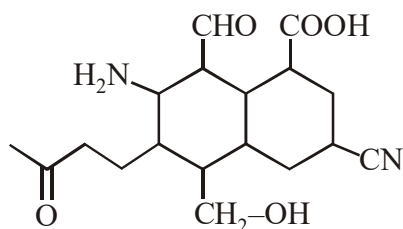
-									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

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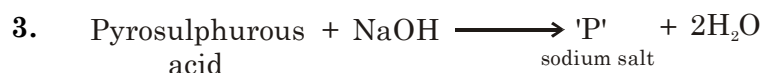
1. How many 1° carbon present in the following?



2. How many functional groups are present in the following



Space for Rough Work



Find the value of expression $\frac{(x+y)}{5}$ for salt 'P'

where

x = total number of bond angle

y = maximum number of equal bond length

4. Assuming z axis is internuclear axis, find total number of combination of atomic orbitals, in which nodal plane of formed bond passes through internuclear axis.

$p_x + p_x, p_z + p_z, d_{xy} + d_{xy}, d_{yz} + d_{yz}, d_{x^2-y^2} + d_{x^2-y^2}, p_y + d_{yz}, p_x + d_{xz}, d_{z^2} + d_{z^2}, p_z + d_{z^2}$

5. With what minimum pressure (in MPa) must a given volume of nitrogen gas ($C_{v,m} = 5/2 R$), originally at 300K and 0.10 MPa pressure, be adiabatically compressed in order to raise its temperature to 600K.

6. One gm sample of NH_4NO_3 (s) is decomposed in a bomb calorimeter (constant volume). The temperature of the calorimeter system rises by 6K. If the heat capacity of system is 1.25 kJ/K, what is the molar enthalpy of decomposition of NH_4NO_3 (s) at 300 K.

[$R = 8.3 \text{ JK}^{-1} \text{ mol}^{-1}$]

Space for Rough Work

PART-3 : MATHEMATICS**SECTION-I(i) : (Maximum Marks : 18)**

- This section contains **SIX** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. A man walks a distance of 5 units starting from origin along the line $y = \sqrt{3}x$ in its increasing direction. Then he turns towards his left and walks a distance of 7 units in a direction parallel to the line $y = -\sqrt{3}x$ and reaches a point P. Then distance of P from Q(10, $2\sqrt{3}$) will be :-
(A) 5 (B) $7\sqrt{3}$ (C) 12 (D) 13
2. The number of solutions of equation $2^{\log_2^2(x^2-3x+3)} + (x^2-3x+3)^{\log_2(x^2-3x+3)} = 2$ is -
(A) 1 (B) 2 (C) 3 (D) 4
3. The radius of the circle passing through the center of incircle of ΔABC and through the end points of BC is given by :
(A) $\frac{a}{2} \cos A$ (B) $\frac{a}{2} \sec \frac{A}{2}$ (C) $\frac{a}{2} \sin A$ (D) $a \sec \frac{A}{2}$

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4. If $a_0 = \cos\theta$ where $\theta \in \left(0, \frac{\pi}{2}\right)$ and $a_{r+1} = \sqrt{\frac{1}{2}(1+a_r)}$ then $\frac{\sqrt{1-a_0^2}}{(a_1 \cdot a_2 \cdot a_3 \dots a_n) \sqrt{1-a_n^2}}$ is equal to
- (A) 2^n (B) $\frac{1}{2^n}$ (C) $2^n \sin \frac{\theta}{2^n}$ (D) $\frac{1}{2^n} \sin \frac{\theta}{2^n}$
5. Let α, β, γ be the roots of $x^3 - 3x^2 + 2x + 4 = 0$ and $y = 1 + \frac{\alpha}{x-\alpha} + \frac{\beta x}{(x-\alpha)(x-\beta)} + \frac{\gamma x^2}{(x-\alpha)(x-\beta)(x-\gamma)}$, then the value of y at $x = 2$ is equal to -
- (A) 1 (B) 2 (C) 4 (D) 8
6. If a, b, c are the sides of the triangle ABC, such that $\left(1 + \frac{b-c}{a}\right)^a \cdot \left(1 + \frac{c-a}{b}\right)^b \cdot \left(1 + \frac{a-b}{c}\right)^c \geq 1$, then triangle ABC must be-
- (A) right angled but not isosceles (B) right angled isosceles
(C) obtuse angled (D) equilateral

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SECTION-I(ii) : (Maximum Marks: 24)

- This section contains **SIX** questions.
 - Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
 - For each question, choose the correct option(s) to answer the question.
 - Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
 - **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.
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7. If $y = |ax^2 + b|x| + c|$ & the line $y = k$ has 8 solutions, where $k \in (0, 4)$, then which of the following is/are always true ($a, b, c \neq 0$) -
- (A) quadratic equation $ax^2 + bx + c = 0$ has roots opposite in sign.
(B) quadratic equation $ax^2 + bx + c = 0$ has both roots positive.
(C) quadratic equation $ax^2 + bx + c = 0$ has both roots of same sign.
(D) $|b^2 - 4ac| > 16|a|$.
8. A straight line passes through (2, 3) and forms a triangle of area S units with positive x & y axes-
- (A) $S_{\min} = 6$
(B) $S_{\min} = 12$
(C) Sum of intercepts on x & y axes has minimum value $5 + 2\sqrt{6}$
(D) Sum of intercepts on x & y axes has minimum value $5 + 6\sqrt{2}$

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9. The solution of the following equation : $2(\sin x - \cos 2x) - \sin 2x(1 + 2\sin x) + 2\cos x = 0$ can be -

(A) $x = 2n\pi ; n \in I$ (B) $x = n\pi + (-1)^n \left(-\frac{\pi}{2}\right) ; n \in I$

(C) $x = n\pi + (-1)^n \frac{\pi}{6} ; n \in I$ (D) $x = n\pi + (-1)^n \frac{\pi}{4} ; n \in I$

10. If $2\tan 10^\circ + \tan 50^\circ = 2x$, $\tan 20^\circ + \tan 50^\circ = 2y$, $2\tan 10^\circ + \tan 70^\circ = 2w$ and $\tan 20^\circ + \tan 70^\circ = 2z$, then which of the following is/are true -

(A) $z > w > y > x$ (B) $w = x + y$
(C) $2y = z$ (D) $z + x = w + y$

11. If $\begin{vmatrix} a_1 + b_1x & a_1x + b_1 & c_1 \\ a_2 + b_2x & a_2x + b_2 & c_2 \\ a_3 + b_3x & a_3x + b_3 & c_3 \end{vmatrix} = 0$, then which can be the possible condition -

(A) $x = 1 \forall a_i, b_i$, where $1 \leq i, j \leq 3$ (B) $x = -1 \forall a_i, b_i$, where $1 \leq i, j \leq 3$

(C) $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$ (D) $x = \pm 2 \forall a_i, b_i$, where $1 \leq i, j \leq 3$

12. Consider two lines $L_1 \equiv 2x - y + 4 = 0$ and $L_2 \equiv x - 2y - 1 = 0$, then which of the following is/are correct -

(A) acute angle bisector of L_1 and L_2 is $x - y + 4 = 0$

(B) obtuse angle bisector of L_1 and L_2 is $x + y + 5 = 0$

(C) acute angle of the lines contains the origin

(D) acute angle between L_1 and L_2 is $\tan^{-1}\left(\frac{7}{4}\right)$

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SECTION-II : (Maximum Marks: 24)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

-									
●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	●	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- Three straight lines are drawn parallel to the sides of a triangle from a point inside the triangle dividing the triangle into 6 regions, three of which are triangles. If area of these triangles are 4, 9, 16 square units, area of original triangle is-
- A finite sequence of positive integers, m_i for $i = 1, 2, 3, \dots, 2008$ are defined so that $m_1 = 1$ and $m_{i+1} = 10m_i + 1, \forall i \geq 1$ How many of these integers are divisible by 3 ?
- Let $f(x)$ be a polynomial function of second degree with leading coefficient 1. If $f(0) = 48$ and $f(3) = f(-3)$, then minimum value of $g(x)$, where $g(x) = (f(x) + f(x+1))$ is -

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4. The sixth term of an A.P. is equal to 2. The value of the common difference 'd' of the A.P. which makes the product $a_1 a_4 a_5$ greatest, is (where $d \geq 1$)
5. Let 't' denotes the value of the product $(1 + a + a^2 + \dots \infty)(1 + b + b^2 + \dots \infty)(1 + c + c^2 + \dots \infty)$ where a,b,c are roots of the equation $12x^3 - 4x^2 - 3x + 1 = 0$. If $y = \log_{p+t} (3x^2 + 2x - (2t + p))$ is defined $\forall x \in \mathbb{R}$, then number of possible integral values of 'p' is
6. The perimeter of a triangle ABC is 6 times the arithmetic mean of the sines of its angles. If the side a is 1, b is $\frac{\sqrt{3}-1}{\sqrt{2}}$ and $\cos A > 0$, then find the value of $\left[100 \left(\sum \sin^2 2A + 2 \prod \sin 2A \sin 2B \right) \right]$, where [.] denotes greatest integer function.

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QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	Single correct option	6	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	–1 In all other cases	18
I(ii)	One or more correct option(s)	6	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	–2 In all other cases	24
II	Numerical Value Type (Up to second decimal place)	6	+4 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	24

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator